

Swarm Behavior Classification Using Machine Learning

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CSCI 460 – Applied Machine Learning

Introduction:

Swarm behavior refers to coordinated movement in multi-agent systems such as birds, fish, and robotics. This project uses machine learning to classify whether a swarm exhibits coordinated behavior.

The dataset contains about 72,000 samples and over 2,400 features describing agent interactions such as position, velocity, alignment, separation, and cohesion.

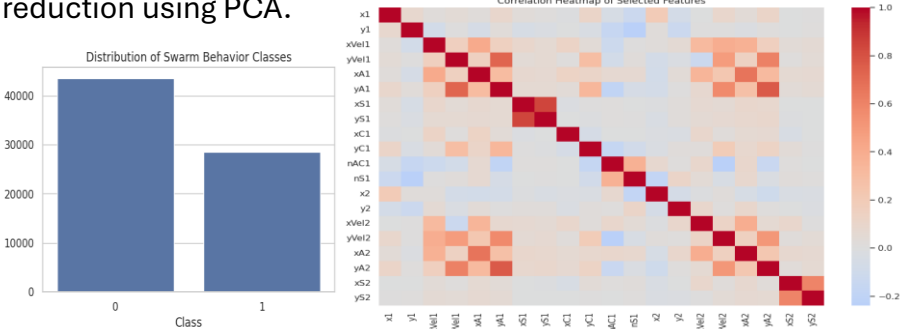
Objective:

To compare multiple machine learning models and determine which performs best for high-dimensional swarm classification.

Exploratory Data Analysis revealed:

- Moderate class imbalance
- High feature dimensionality (~2400 features)
- Strong correlation between features (redundancy)
- Wide variation in feature distributions

These findings motivated scaling and dimensionality reduction using PCA.



Data Preprocessing:

- Converted all features to numeric
- Checked for missing values
- StandardScaler applied to normalize data

Dimensionality Reduction:

- PCA reduced features from ~2400 → 50
- Improved efficiency and reduced overfitting

Models Implemented:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Support Vector Machine (SVM)
- Random Forest
- XGBoost

Hyperparameter Tuning:

- GridSearchCV with cross-validation

Evaluation Metrics:

- Accuracy, Precision, Recall, F1-score
- F1-score emphasized due to class imbalance

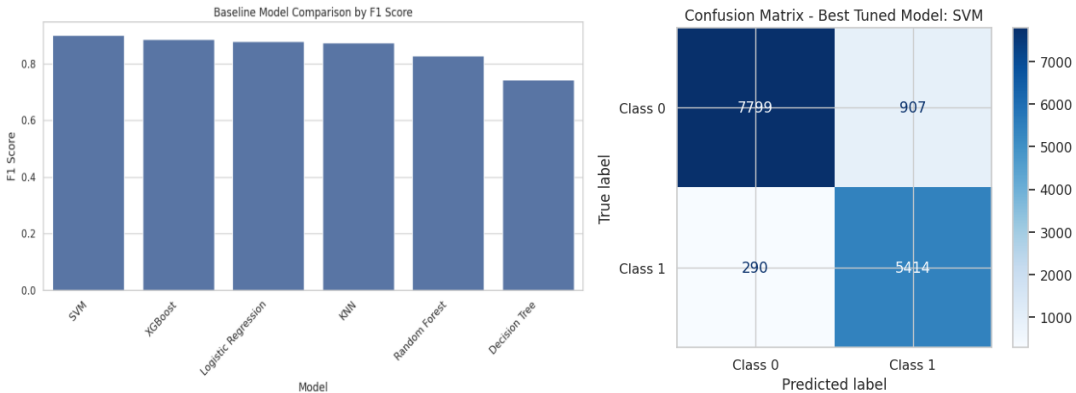
Conclusion:

- Machine learning successfully classified swarm behavior
- SVM achieved the highest performance (F1 ≈ 0.90)
- PCA significantly improved efficiency and reduced overfitting

Model Performance (F1 Score):

SVM: 0.9008 (Best)
XGBoost: 0.8854
Logistic Regression: 0.8776
KNN: 0.8739
Random Forest: 0.8270
Decision Tree: 0.7427

SVM achieved the best performance due to its ability to handle high-dimensional feature spaces.



Future Work:

- Explore feature selection instead of PCA
- Further tune XGBoost and SVM
- Evaluate additional models